

Musa Ibne Mannan

Dallas, Texas 75252 | (737) 221-1884 | musa314@utdallas.edu
musamannan.com | linkedin.com/in/musa314 | ORCID 0009-0005-5579-6899

Summary

Mechanical engineering Ph.D. candidate with five years of combined research and field engineering experience in failure analysis, finite element analysis, and experimental characterization. Built and commissioned high-vacuum CVD and Raman spectroscopy systems; performed root cause analysis on 120+ design faults across refinery and boiler projects, saving approximately 30,000 man-hours. Proficient in Abaqus, Python, MATLAB, SolidWorks, and materials characterization (SEM, EDX, XPS, XRD). Seeking forensic mechanical engineering, failure analysis, and technical consulting roles.

Education

Doctor of Philosophy (Ph.D.) in Mechanical Engineering Aug 2023 – Present
Erik Jonsson School of Engineering and Computer Science, The University of Texas at Dallas, Richardson, TX
Concentration: Manufacturing and Design Innovations GPA: 3.62

Master of Science in Mechanical and Manufacturing Engineering (ABET accredited) Aug 2021 – May 2023
Ingram School of Engineering, Texas State University, San Marcos, TX GPA: 3.75

Technical Skills

Failure Analysis and Testing: Root cause analysis, fracture surface examination (SEM, EDAX), mechanical testing (tensile, flexural, Izod impact; MTS 810), thermal analysis (TGA, DSC, TMA), design review, field error resolution.

Simulation and CAD: Abaqus (transient heat transfer, thermal stress, plasticity from true stress-strain data, mesh convergence, 2D plane stress and 3D solid elements), SolidWorks, AutoCAD, CATIA, Mastercam, basic CFD.

Programming and Data Analysis: Python (data analysis pipelines, Random Forest, XGBoost), MATLAB (numerical methods, custom FEM implementation, Gauss quadrature, multi-objective genetic algorithms).

Experimental and Fabrication: High-vacuum CVD system design and commissioning, pulsed laser deposition, Raman spectroscopy, XPS, XRD, SEM/EDX, ellipsometry, Hall measurement, FDM/SLA/Markforged 3D printing, twin-screw extrusion, CNC machining (HAAS VF-3), laser cutting, waterjet, TIG and arc welding, mill and lathe.

Professional: Technical report writing, publication-quality figures and presentations, EHS and OSHA compliance, procurement and inventory management, Adobe Creative Suite.

Professional Experience

Research Assistant and Lab In-Charge Jun 2024 – Present
Leem Research Group, The University of Texas at Dallas, Richardson, TX

- Independently designed, assembled from the ground up, troubleshoot, and operated high-vacuum CVD furnaces, a hydrogen generator, and integrated Raman spectroscopy instrumentation, then optimized the resulting process.
- Led complete laboratory build-out including procurement, equipment installation, safety compliance, and inventory management while running concurrent computational and experimental programs.
- Applied Python-based data analysis pipelines and ensemble machine learning (Random Forest, XGBoost) to optimize multi-parameter mechanical system performance and materials synthesis processes.

Graduate Teaching Associate (Instructor of Record) Aug 2025 – Present
The University of Texas at Dallas, Richardson, TX

- Serve as sole instructor for ECS 1100, responsible for curriculum delivery, mentoring, and assessment of 130 first-year engineering students; develop and deliver technical lectures on engineering fundamentals including mechanics and design.

Assistant and Site Engineer Sep 2018 – Jun 2019
Washillah Engineering Services (Partex Petro Ltd. and CUFL projects), Bangladesh

- Led on-site teams of 28 to 35 engineers and contractors for large-scale mechanical and piping projects (condensate refinery and boiler re-tubing); enforced EHS and OSHA standards and managed field operations under tight deadlines.
- Performed root cause analysis and resolved over 120 design and implementation errors in mechanical systems and structures, saving approximately 30,000 man-hours and \$1M while maintaining quality and schedule.
- Coordinated cross-functional teams, client interfaces, material logistics, and quality control in high-stakes industrial field environments.

Graduate Instructional Assistant

Aug 2023 – May 2024

The University of Texas at Dallas, Richardson, TX

- Graded assignments and exams and tutored 140+ students in Kinematics and Dynamics (MECH 3350) and Mechanics of Materials (MECH 2320).
- Delivered substitute lectures and assisted students with computational modeling, FEA, and CAD troubleshooting.

Graduate Instructional Assistant

Sep 2021 – May 2023

Texas State University, San Marcos, TX

- Mentored three senior design teams (15 students) to completion on industry-sponsored mechanical engineering capstone projects; provided hands-on guidance in CAD, FEA, prototyping, and mechanical system design.

Selected Projects

- **Machine Gun Barrel Thermal and Stress Failure Analysis (Abaqus FEA):** Calculated firing pressure analytically at approximately 130.5 MPa, then ran transient heat transfer and von Mises stress analysis under sustained high-rate firing; evaluated AISI 4140, AK Steel 316L, and ASTM A656 Grade 7 for thermal gradients, stress concentration, and failure risk.
- **Cantilever Beam FEA and Mesh Convergence Study (Abaqus, MATLAB):** Analyzed a 10 m cantilever under 2,000 N shear using CPS4, CPS8, and C3D8 elements across three mesh densities; 3D deflection error fell from 84% to 22%, 2D mid-span stress error from 94% to 8.6%. Verified against custom MATLAB code using 2x2 Gauss quadrature.
- **Biocomposite Characterization and Failure Diagnosis:** Designed, extruded, and 3D-printed silane-treated natural fiber composites; used SEM/EDAX, tensile and flexural testing, and thermal analysis to diagnose interfacial adhesion issues, fiber-size effects, and resulting mechanical failure modes.
- **rGO Thin-Film Processing and Characterization:** Independently designed and executed pulsed laser deposition and annealing experiments; performed full characterization (XPS, Raman, XRD, ellipsometry, Hall measurement, low-temperature transport).

Publications

1. **Musa Ibne Mannan**, Varin Kansal, Premesh N. Mistry, Victoria Estrada, Juyoung Leem. “Hardware-Agnostic 2D Material Synthesis via Zero-Shot Machine Learning Translation.” *Advanced Materials* (Wiley). Under review.
2. **Musa Ibne Mannan**, Premesh Mistry, Varin Kansal, Victoria Estrada, Juyoung Leem. “AI-Enabled CVD Synthesis, Quality Control, and Autonomous Manufacturing of 2D Materials.” *npj Advanced Manufacturing* (Springer Nature), 2026. DOI: 10.1038/s44334-026-00109-5.
3. Wasi Shadman, **Musa Mannan**, et al. “Electrochemical Detection of tRNA-Derived Fragments (tRFs) Based on Multifunctionalized Carbon Surface.” *Surfaces and Interfaces* (Elsevier). Under review.
4. MSHK Tushar, **MI Mannan**, A Azmain. “Exhaust Gas After Treatment using Air Preheating and Selective Catalytic Reduction by Urea to Reduce NOx in Diesel Engine.” *Heliyon*, 11(1), 2025.
5. KM Jitendra Tate, **Musa Ibne Mannan**, et al. “Thermal and Structural Properties of Surface Treated Ragweed Fiber Reinforced Biodegradable Polylactic Acid Biocomposites.” *International Journal of Composite and Constituent Materials*, 8(2), 2022.
6. S Panta, **MI Mannan**, et al. “Comparison of Mechanical and Morphological Properties of Amino and Glymo Silane Treated Ragweed-PLA Bio-Composites.” *CAMX 2022 Proceedings*, Anaheim, CA, 2022.

Additional Experience

Emergency Room Scribe Workflow Innovation, Augmedix

Mar 2020 – Jul 2021

Designed and piloted the first remote ER medical scribing program; built the training curriculum from scratch and implemented a real-time documentation workflow across multi-site teams. Featured in *Business Insider*.

Medical Scribe, Augmedix

Jul 2019 – Jul 2021

Delivered real-time documentation for 4,000+ patients across multiple medical disciplines under timed, high-pressure conditions.

Novelist (Bangla)

Feb 2016 – Present

Author of 17 published crime novels, written to deadline alongside full-time engineering and research work.